Homework 2 Pseudocode

Bayou City Family Health Patient Registration Form

- Name: Brian Nwabuoku
- Class: MIS 7375 Transaction Processing Systems I
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Main Program

MAIN

DO BEGINNING
DO THE MIDDLE
DO THE END
END MAIN

DO BEGINNING

DISPLAY THE HTML FORM (homework2.html)
SHOW TODAYS DATE IN THE BANNER
LOAD THE STATE LIST FROM homework2-states.js INTO THE STATE BOX
    PUT "Select" FIRST WITH NO VALUE SO THE USER HAS TO PICK ONE
SHOW THE SLIDER VALUE NEXT TO THE BAR
SET-CONDITIONS (onblur, onchange and oninput on the fields, onclick on Review)
END BEGINNING

DO MIDDLE (javascript, homework2.js)

IF THE USER ENTERED AND THEN LEFT A FIELD
    CALL SHOW-RESULT FOR THAT FIELD
ENDIF
IF THE USER IS TYPING IN A PASSWORD BOX
    CALL SHOW-RESULT FOR THAT PASSWORD BOX
ENDIF
IF THE USER MOVED THE SLIDER
    CALL SHOW-SLIDER-VALUE
ENDIF
IF THE REVIEW BUTTON WAS CHOSEN
    CALL REVIEW-FORM
ENDIF

```
IF THE SUBMIT BUTTON WAS CHOSEN
        LET THE BROWSER CHECK REQUIRED FIELDS AND PATTERNS
        IF EVERYTHING PASSES
                DISPLAY THANK YOU PAGE
        ENDIF
ENDIF
IF THE RESET BUTTON WAS CHOSEN
        CLEAR THE FORM
ENDIF
END MIDDLE
```

THE-END

```
SHOW THE THANK YOU PAGE (thankyou.html)
GIVE A LINK BACK TO THE FORM
END THE-END
```

Helper Modules

GET-FIELD

```
FIND THE FIELD IN THE FORM BY ITS NAME (NOT ITS ID)
RETURN THE FIELD
END GET-FIELD
```

GET-VALUE

```
CALL GET-FIELD
TRIM THE EXTRA SPACES OFF THE ENDS
PUT THE TRIMMED VALUE BACK IN THE BOX
RETURN THE VALUE
END GET-VALUE
```

SHOW-RESULT (field name)

```
CALL THE CHECK FOR THAT FIELD
PUT WHAT IT RETURNS IN THE ERROR SPOT NEXT TO THE FIELD (FOUND BY ID)
IF THIS IS PASSWORD 1 AND PASSWORD 2 IS NOT EMPTY
        CALL SHOW-RESULT FOR PASSWORD 2
ENDIF
END SHOW-RESULT
```

SHOW-SLIDER-VALUE

DISPLAY THE SLIDER NUMBER AS "N out of 10" NEXT TO THE BAR
END SHOW-SLIDER-VALUE

Field Checks

EVERY CHECK

CHECK ONLY ITS OWN FIELD
RETURN THE ERROR MESSAGE IF SOMETHING IS WRONG
RETURN NOTHING IF THE FIELD IS GOOD
LEAVE THE ERROR SPOT TO SHOW-RESULT

CHECK-FIRST-NAME

IF FIRST NAME IS EMPTY
    RETURN "First name is required"
ELSE IF IT HAS ANYTHING BUT LETTERS APOSTROPHES AND DASHES OR IS OVER 30 CHARS
    RETURN "Letters, apostrophes and dashes only"
ENDIF
END CHECK-FIRST-NAME

CHECK-MIDDLE-INIT

IF MIDDLE INITIAL IS NOT EMPTY AND IS NOT ONE LETTER
    RETURN "Middle initial has to be one letter"
ENDIF
END CHECK-MIDDLE-INIT

CHECK-LAST-NAME

IF LAST NAME IS EMPTY
    RETURN "Last name is required"
ELSE IF IT HAS ANYTHING BUT LETTERS APOSTROPHES DASHES OR NUMBERS 2 TO 5
    RETURN "Letters, apostrophes, dashes and numbers 2 to 5 only"
ENDIF
END CHECK-LAST-NAME

CHECK-DOB

IF DATE OF BIRTH IS EMPTY

```
        RETURN "Date of birth is required"
ELSE IF IT IS NOT MM/DD/YYYY
        RETURN "Please use MM/DD/YYYY"
ELSE
        SPLIT IT INTO MONTH DAY AND YEAR
        IF THAT DAY DOES NOT EXIST (LIKE 02/30)
                RETURN "That date doesn't exist"
        ENDIF
        GET TODAYS DATE (SAME DATE SHOWN IN THE BANNER)
        SET OLDEST DATE TO TODAY MINUS 120 YEARS
        IF DATE OF BIRTH IS AFTER TODAY
                RETURN "Can't be in the future"
        ELSE IF DATE OF BIRTH IS BEFORE THE OLDEST DATE
                RETURN "Can't be more than 120 years ago"
        ENDIF
ENDIF
END CHECK-DOB
```

CHECK-SSN

```
IF SSN IS EMPTY
        RETURN "Social Security number is required"
ELSE IF IT IS NOT 9 DIGITS (DASHES OK)
        RETURN "SSN needs 9 digits"
ENDIF
END CHECK-SSN
```

CHECK-ADDRESS (address 1, address 2 and city)

```
IF ADDRESS 1 OR CITY IS EMPTY
        RETURN "Required"
ELSE IF ADDRESS 1 OR CITY IS LESS THAN 2 OR MORE THAN 30 CHARS
        RETURN "Needs 2 to 30 characters"
ENDIF
IF ADDRESS 2 IS NOT EMPTY AND IS LESS THAN 2 OR MORE THAN 30 CHARS
        RETURN "Needs 2 to 30 characters or leave it blank"
ENDIF
END CHECK-ADDRESS
```

CHECK-STATE

```
IF STATE IS STILL "Select"
        RETURN "Please pick a state"
ENDIF
```

END CHECK-STATE

CHECK-ZIP

IF ZIP IS EMPTY
    RETURN "Zip code is required"
ELSE IF IT IS NOT 5 DIGITS OR 5 DIGITS DASH 4 DIGITS
    RETURN "Zip code needs 5 digits"
ELSE
    CUT IT TO THE FIRST 5 DIGITS
    REDISPLAY THE SHORTER ZIP IN THE BOX
ENDIF
END CHECK-ZIP

CHECK-PHONE

IF PHONE IS EMPTY
    RETURN "Phone number is required"
ELSE IF IT IS NOT 10 DIGITS
    RETURN "Phone number needs 10 digits"
ELSE
    TAKE OUT EVERYTHING THAT IS NOT A DIGIT
    REDISPLAY IT AS 000-000-0000
ENDIF
END CHECK-PHONE

CHECK-EMAIL

IF EMAIL IS EMPTY
    RETURN "Email address is required"
ELSE IF IT IS NOT name@domain.tld
    RETURN "Email needs to look like name@domain.com"
ENDIF
END CHECK-EMAIL

CHECK-USER-ID

IF USER ID IS EMPTY
    RETURN "User ID is required"
ELSE IF IT HAS A SPACE
    RETURN "No spaces"
ELSE IF IT STARTS WITH A NUMBER
    RETURN "Can't start with a number"
ELSE IF IT IS NOT 5 TO 30 LETTERS NUMBERS UNDERSCORES OR DASHES

```
        RETURN "5 to 30 letters, numbers, underscores or dashes"
ELSE
        CHANGE IT TO LOWERCASE
        REDISPLAY THE LOWERCASE USER ID IN THE BOX
ENDIF
END CHECK-USER-ID
```

CHECK-PASSWORD-1 (runs as the user types)

```
IF PASSWORD IS EMPTY
        RETURN "Password is required"
ENDIF
START A LIST OF WHAT IS MISSING
IF LESS THAN 8 OR MORE THAN 30 CHARS ADD "8 to 30 characters"
IF NO UPPERCASE LETTER ADD "an uppercase letter"
IF NO LOWERCASE LETTER ADD "a lowercase letter"
IF NO NUMBER ADD "a number"
IF NO SPECIAL CHARACTER ADD "a special character"
IF IT HAS A DOUBLE QUOTE
        RETURN "No double quotes"
ELSE IF THE LIST IS NOT EMPTY
        RETURN "Still needs" AND THE LIST
ELSE IF IT HAS THE USER ID IN IT
        RETURN "Can't include your user ID"
ELSE IF IT HAS THE FIRST OR LAST NAME IN IT
        RETURN "Can't include your name"
ENDIF
END CHECK-PASSWORD-1
```

CHECK-PASSWORD-2 (runs as the user types)

```
IF PASSWORD 2 IS EMPTY
        RETURN "Please type your password again"
ELSE IF PASSWORD 2 DOES NOT EQUAL PASSWORD 1
        RETURN "Passwords do not match"
ENDIF
END CHECK-PASSWORD-2
```

CHECK-RADIO (gender, vaccinations, insurance)

```
IF NO BUTTON IN THE GROUP IS CHOSEN
        RETURN "Please answer this question"
ENDIF
END CHECK-RADIO
```

CHECK-SYMPTOMS

```
IF THE TEXT AREA HAS A DOUBLE QUOTE
        RETURN "Please leave out double quotes"
ENDIF
END CHECK-SYMPTOMS
```

Review

REVIEW-FORM

```
CALL EVERY CHECK AND KEEP THE RESULTS
SHOW EACH RESULT IN THE ERROR SPOT NEXT TO ITS FIELD
DISPLAY HEADING "PLEASE REVIEW YOUR INFORMATION"
LOOP THROUGH EVERY LINE OF THE REVIEW UNTIL NONE LEFT
        DISPLAY THE FIELD LABEL FROM THE PAGE (NOT THE VARIABLE NAME)
        CHECK THE KIND OF DATA
                CASE OF NAME
                        DISPLAY FIRST MI AND LAST TOGETHER
                END CASE
                CASE OF ADDRESS
                        DISPLAY ADDRESS 1, ADDRESS 2 IF THERE IS ONE, CITY STATE
AND ZIP
                END CASE
                CASE OF SSN
                        DISPLAY ONLY THE LAST 4 DIGITS
                END CASE
                CASE OF A RADIO BUTTON
                        DISPLAY THE WORDING OF THE BUTTON THAT WAS CHOSEN
                END CASE
                CASE OF A CHECKBOX
                        IF THIS ITEM WAS CHECKED
                                DISPLAY "Y"
                        ELSE
                                DISPLAY "N"
                        ENDIF
                END CASE
                CASE OF THE SLIDER
                        DISPLAY "N out of 10"
                END CASE
                CASE OF THE PASSWORD
                        DISPLAY STARS ONLY
                END CASE
```

```
                CASE OF BUTTONS
                        DO NOTHING
                END CASE
                DEFAULT CASE
                        DISPLAY VALUE
                END CASE
        IF THE CHECK FOR THIS LINE PASSED
                DISPLAY "pass"
        ELSE
                DISPLAY "ERROR" AND THE MESSAGE
        ENDIF
END LOOP
COUNT THE ERRORS
IF NO ERRORS
        DISPLAY "Everything looks good. Click Submit"
ELSE
        DISPLAY "Please fix the errors and click Review again"
ENDIF
RETURN THE NUMBER OF ERRORS
END REVIEW-FORM
```